FAIRCHILD SEMICONDUCTOR®

March 2009

FDG6332C_F085

20V N & P-Channel PowerTrench® MOSFETs

Features

- **Q1** 0.7 A, 20V. $R_{DS(ON)}$ = 300 mΩ @ V_{GS} = 4.5 V
  $R_{DS(ON)}$ = 400 mΩ @ V_{GS} = 2.5 V
- **Q2** –0.6 A, –20V. $R_{DS(ON)}$ = 420 mΩ @ V_{GS} = –4.5 V
  $R_{DS(ON)}$ = 630 mΩ @ V_{GS} = –2.5 V
- Low gate charge
- High performance trench technology for extremely low $R_{DS(ON)}$
- SC70-6 package: small footprint (51% smaller than SSOT-6); low profile (1mm thick)
- Qualified to AEC Q101
- RoHS Compliant

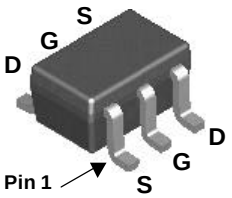
General Description

The N & P-Channel MOSFETs are produced using Fairchild Semiconductor's advanced PowerTrench process that has been especially tailored to minimize on-state resistance and yet maintain superior switching performance.

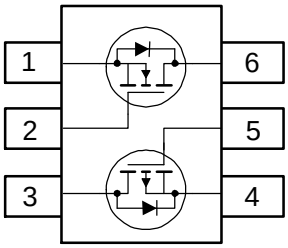
These devices have been designed to offer exceptional power dissipation in a very small footprint for applications where the bigger more expensive TSSOP-8 and SSOP-6 packages are impractical.

Applications

- DC/DC converter
- Load switch
- LCD display inverter



SC70-6



Complementary

Absolute Maximum Ratings

T_A=25°C unless otherwise noted

Symbol	Parameter		Q1	Q2	Units
V_{DSS}	Drain-Source Voltage		20	–20	V
V_{GSS}	Gate-Source Voltage		±12	±12	V
I_D	Drain Current – Continuous	(Note 1)	0.7	–0.6	A
	– Pulsed		2.1	–2	
P_D	Power Dissipation for Single Operation	(Note 1)	0.3		W
T_J, T_{STG}	Operating and Storage Junction Temperature Range		–55 to +150		°C

Thermal Characteristics

$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	(Note 1)	415	°C/W

Package Marking and Ordering Information

Device Marking	Device	Reel Size	Tape width	Quantity
.32	FDG6332C_F085	7"	8mm	3000 units

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Electrical Characteristics

T_A = 25°C unless otherwise noted

Symbol	Parameter		Test Conditions	Min	Typ	Max	Units
Off Characteristics							
BV_{DSS}	Drain–Source Breakdown Voltage		V_{GS} = 0 V, I_D = 250 μA V_{GS} = 0 V, I_D = –250 μA	**Q1** **Q2**	20 –20			V
$\frac{\Delta BV_{DSS}}{\Delta T_J}$	Breakdown Voltage Temperature Coefficient		I_D = 250 μA,Ref. to 25°C I_D = –250 μA,Ref. to 25°C	**Q1** **Q2**		14 –14		mV/°C
I_{DSS}	Zero Gate Voltage Drain Current		V_{DS} = 16 V, V_{GS} = 0 V V_{DS} = –16 V, V_{GS} = 0 V	**Q1** **Q2**			1 –1	μA
I_{GSSF} /I_{GSSR}	Gate–Body Leakage, Forward		V_{GS} = ± 12 V, V_{DS} = 0 V				±100	nA
I_{GSSF} /I_{GSSR}	Gate–Body Leakage, Reverse		V_{GS} = ± 12V , V_{DS} = 0 V				±100	nA
On Characteristics (Note 2)							
$V_{GS(th)}$	Gate Threshold Voltage	**Q1**	V_{DS} = V_{GS}, I_D = 250 μA	0.6	1.1	1.5	V
		Q2	V_{DS} = V_{GS}, I_D = –250 μA	-0.6	–1.2	–1.5	
$\frac{\Delta V_{GS(th)}}{\Delta T_J}$	Gate Threshold Voltage Temperature Coefficient	**Q1** **Q2**	I_D = 250 μA,Ref. To 25°C I_D = –250 μA,Ref. to 25°C		–2.8 3		mV/°C
$R_{DS(on)}$	Static Drain–Source On–Resistance	**Q1**	V_{GS} = 4.5 V, I_D =0.7 A V_{GS} = 2.5 V, I_D =0.6 A V_{GS} = 4.5 V, I_D =0.7A,T_J=125°C		180 293 247	300 400 442	mΩ
		Q2	V_{GS} = –4.5 V, I_D = –0.6 A V_{GS} = –2.5 V, I_D = –0.5 A V_{GS}=–4.5 V, I_D =–0.6 A,T_J=125°C		300 470 400	420 630 700	
g_{FS}	Forward Transconductance	**Q1**	V_{DS} = 5 V I_D = 0.7 A		2.8		S
		Q2	V_{DS} = –5 V I_D = –0.6A		1.8		
$I_{D(on)}$	On–State Drain Current	**Q1**	V_{GS} = 4.5 V, V_{DS} = 5 V	1			A
		Q2	V_{GS} = –4.5 V, V_{DS} = –5 V	–2			
Dynamic Characteristics							
C_{iss}	Input Capacitance	**Q1**	V_{DS}=10 V, V_{GS}= 0 V, f=1.0MHz		113		pF
		Q2	V_{DS}=–10 V, V_{GS}= 0 V, f=1.0MHz		114		
C_{oss}	Output Capacitance	**Q1**	V_{DS}=10 V, V_{GS}= 0 V, f=1.0MHz		34		pF
		Q2	V_{DS}=–10 V, V_{GS}= 0 V, f=1.0MHz		24		
C_{rss}	Reverse Transfer Capacitance	**Q1**	V_{DS}=10 V, V_{GS}= 0 V, f=1.0MHz		16		pF
		Q2	V_{DS}=–10 V, V_{GS}= 0 V, f=1.0MHz		9		
Switching Characteristics (Note 2)							
$t_{d(on)}$	Turn–On Delay Time	**Q1**	For **Q1**: V_{DS} =10 V, I_D= 1 A V_{GS}= 4.5 V, R_{GEN} = 6 Ω For **Q2**: V_{DS} =–10 V, I_D= –1 A V_{GS}= –4.5 V, R_{GEN} = 6 Ω		5	10	ns
		Q2			5.5	11	
t_r	Turn–On Rise Time	**Q1**			7	15	ns
		Q2			14	25	
$t_{d(off)}$	Turn–Off Delay Time	**Q1**			9	18	ns
		Q2			6	12	
t_f	Turn–Off Fall Time	**Q1**			1.5	3	ns
		Q2			1.7	3.4	
Q_g	Total Gate Charge	**Q1**	For **Q1**: V_{DS} =10 V, I_D= 0.7 A V_{GS}= 4.5 V, R_{GEN} = 6 Ω For **Q2**: V_{DS} =–10 V, I_D= –0.6 A V_{GS}= –4.5 V, R_{GEN} = 6 Ω		1.1	1.5	nC
		Q2			1.4	2	
Q_{gs}	Gate–Source Charge	**Q1**			0.24		nC
		Q2			0.3		
Q_{gd}	Gate–Drain Charge	**Q1**			0.3		nC
		Q2			0.4		

Electrical Characteristics

T_A = 25°C unless otherwise noted

Symbol	Parameter		Test Conditions		Min	Typ	Max	Units
Drain–Source Diode Characteristics and Maximum Ratings								
I_S	Maximum Continuous Drain–Source Diode Forward Current			**Q1**			0.25	A
				Q2			–0.25	
V_{SD}	Drain–Source Diode Forward Voltage	**Q1**	V_{GS} = 0 V, I_S = 0.25 A (Note 2)			0.74	1.2	V
		Q2	V_{GS} = 0 V, I_S = –0.25 A (Note 2)			–0.77	–1.2	

Notes:

1. $R_{\theta JA}$ is the sum of the junction-to-case and case-to-ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. $R_{\theta JC}$ is guaranteed by design while $R_{\theta JA}$ is determined by the user's board design. $R_{\theta JA}$ = 415°C/W when mounted on a minimum pad of FR-4 PCB in a still air environment.

2. Pulse Test: Pulse Width < 300µs, Duty Cycle < 2.0%

Typical Characteristics: N-Channel

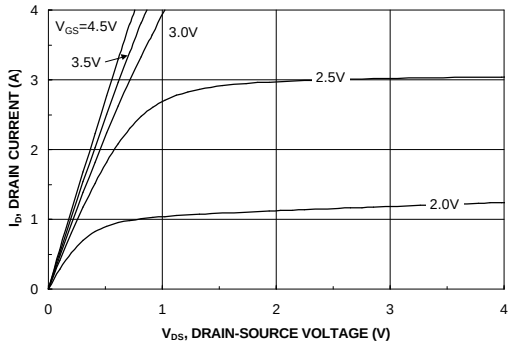


Figure 1. On-Region Characteristics.

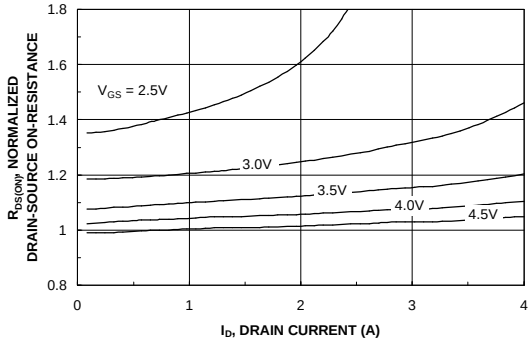


Figure 2. On-Resistance Variation with Drain Current and Gate Voltage.

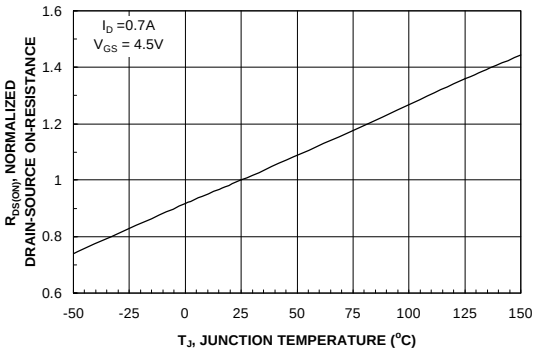


Figure 3. On-Resistance Variation with Temperature.

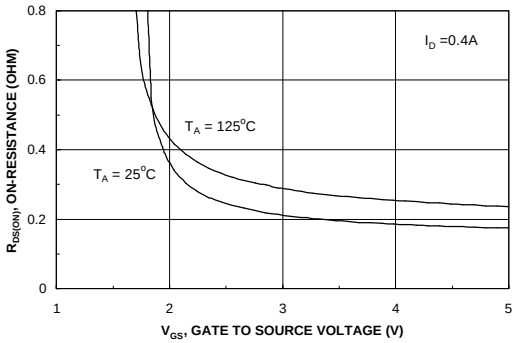


Figure 4. On-Resistance Variation with Gate-to-Source Voltage.

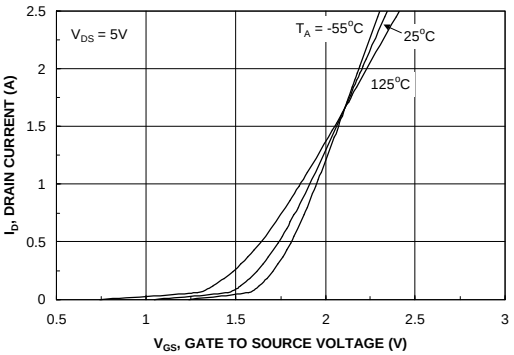


Figure 5. Transfer Characteristics.

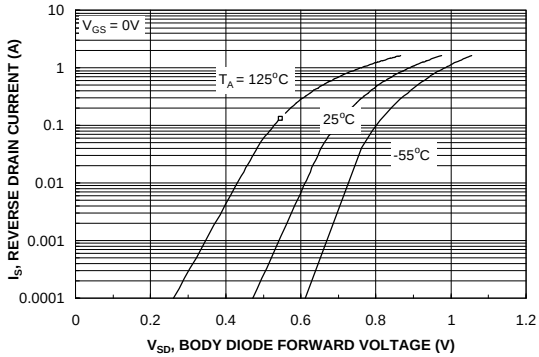


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature.

Typical Characteristics: N-Channel

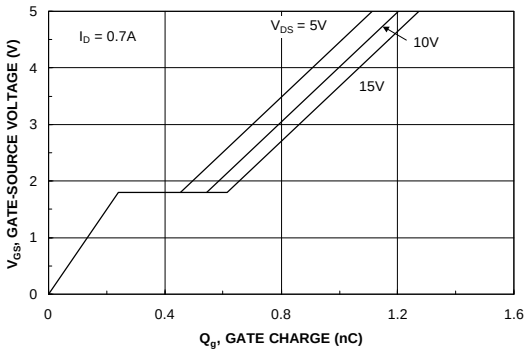


Figure 7. Gate Charge Characteristics.

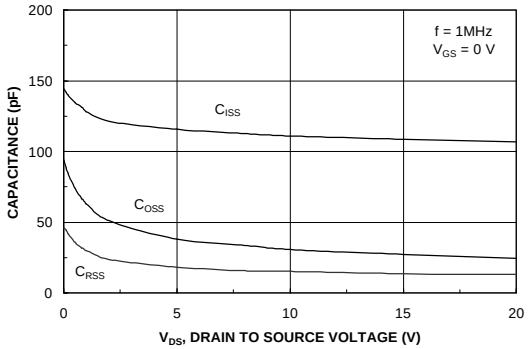


Figure 8. Capacitance Characteristics.

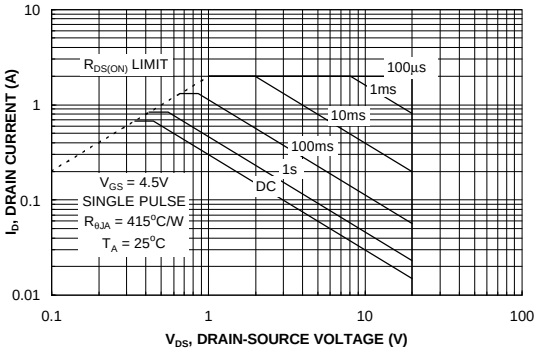


Figure 9. Maximum Safe Operating Area.

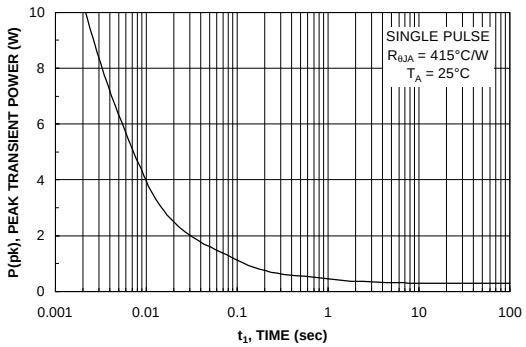


Figure 10. Single Pulse Maximum Power Dissipation.

Typical Characteristics: P-Channel

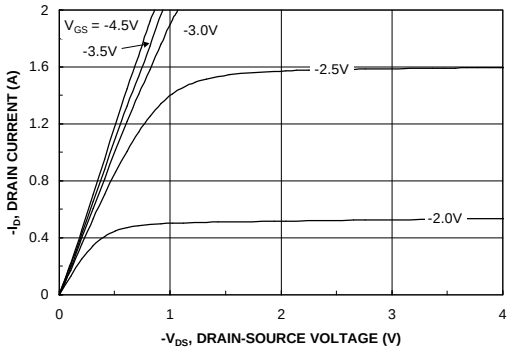


Figure 11. On-Region Characteristics.

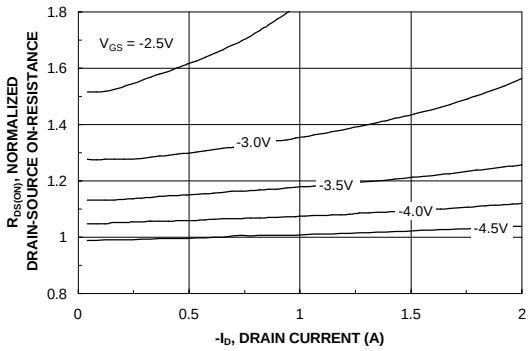


Figure 12. On-Resistance Variation with Drain Current and Gate Voltage.

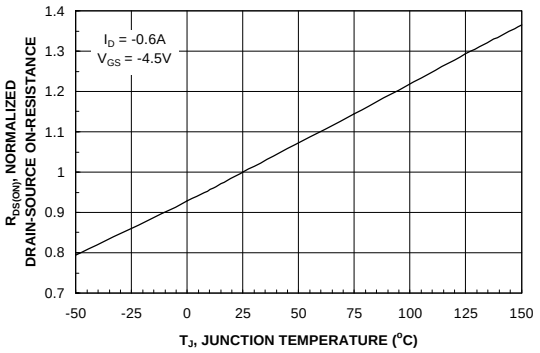


Figure 13. On-Resistance Variation with Temperature.

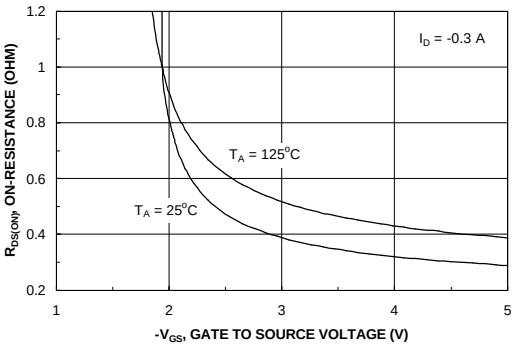


Figure 14. On-Resistance Variation with Gate-to-Source Voltage.

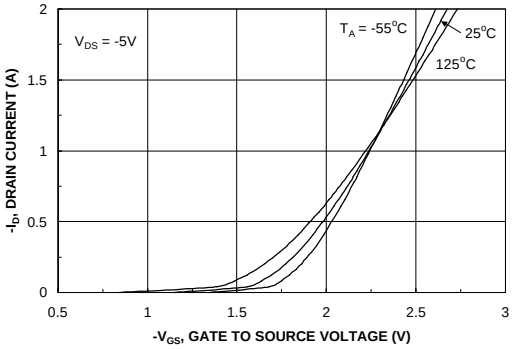


Figure 15. Transfer Characteristics.

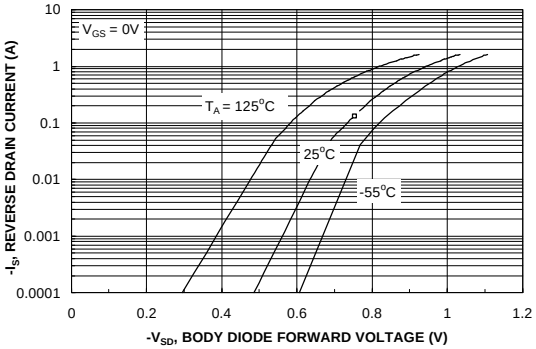


Figure 16. Body Diode Forward Voltage Variation with Source Current and Temperature.

Typical Characteristics: P-Channel

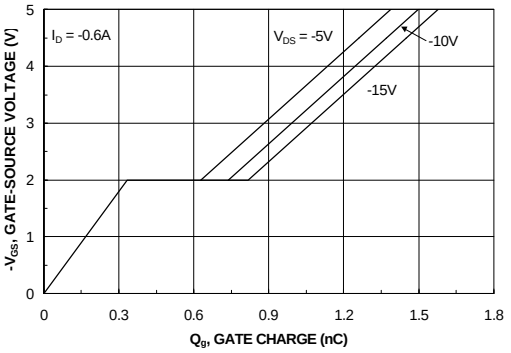


Figure 17. Gate Charge Characteristics.

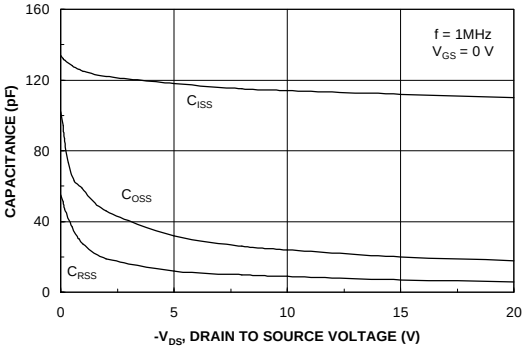


Figure 18. Capacitance Characteristics.

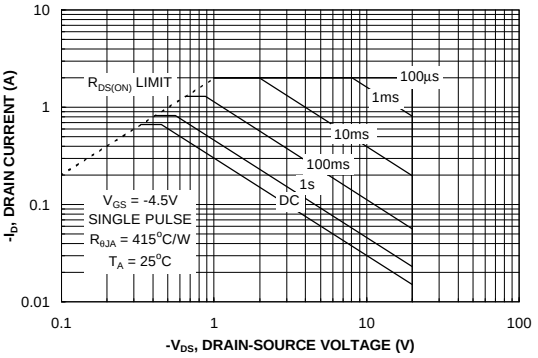


Figure 19. Maximum Safe Operating Area.

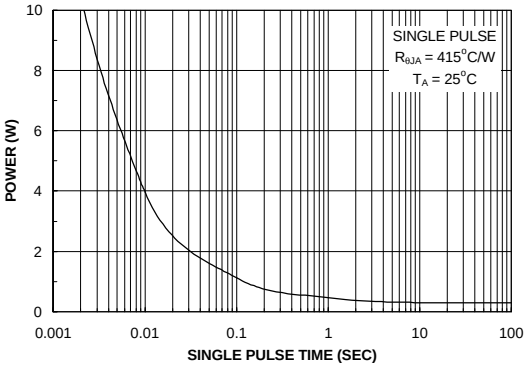


Figure 20. Single Pulse Maximum Power Dissipation.

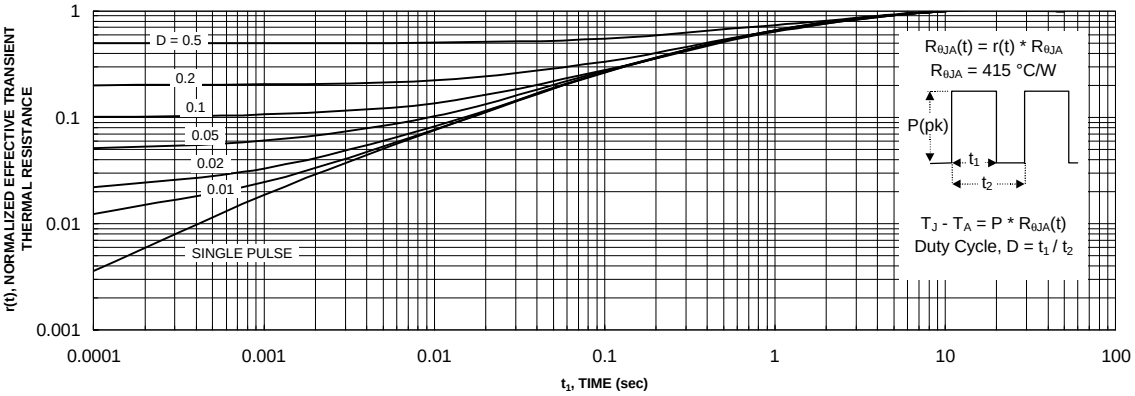
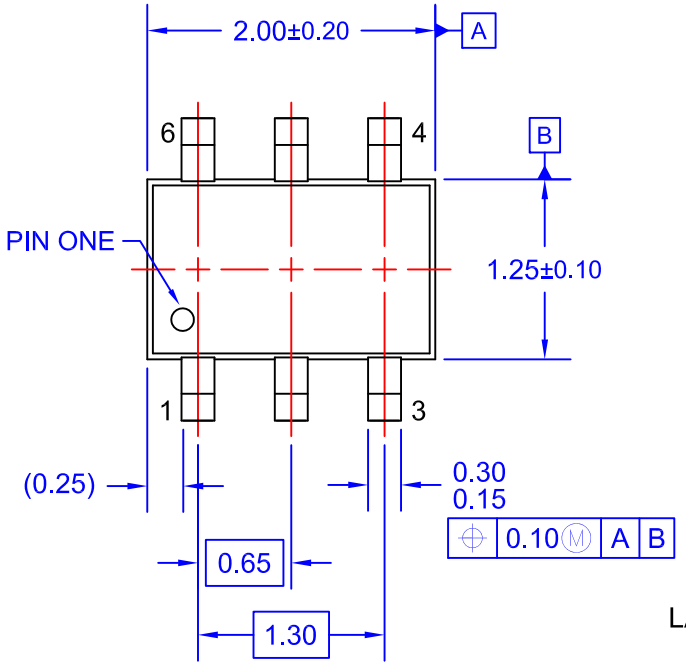
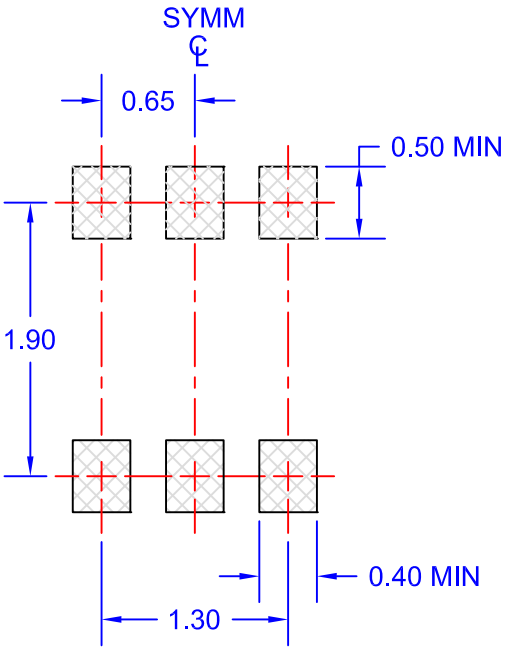


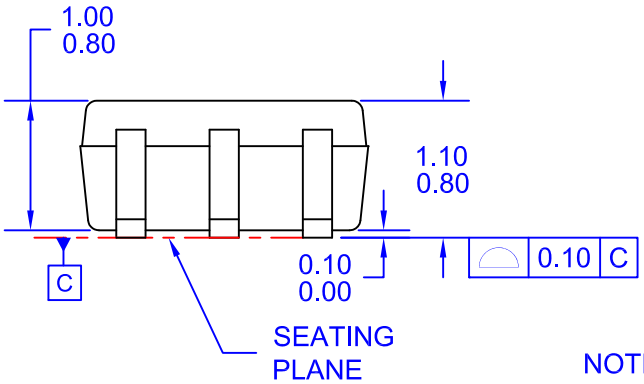
Figure 21. Transient Thermal Response Curve.

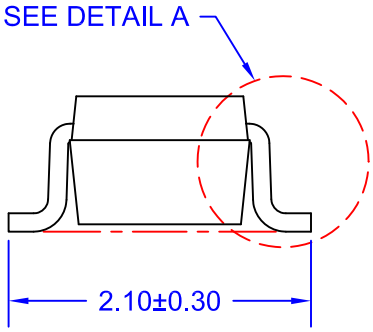
Thermal characterization performed using the conditions described in Note 1.
Transient thermal response will change depending on the circuit board design.





LAND PATTERN RECOMMENDATION





NOTES: UNLESS OTHERWISE SPECIFIED

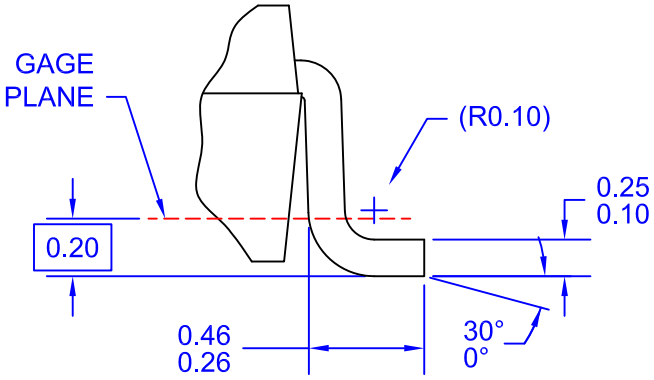
A) THIS PACKAGE CONFORMS TO EIAJ SC-88, 1996.

B) ALL DIMENSIONS ARE IN MILLIMETERS.

C) DIMENSIONS DO NOT INCLUDE BURRS OR MOLD FLASH.

D) DIMENSIONING AND TOLERANCING PER ASME Y14.5M-2009

E) DRAWING FILENAME: MKT-MAA06AREV7



DETAIL A
SCALE: 60X

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ANTI-COUNTERFEITING POLICY

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Counterfeiting of semiconductor parts is a growing problem in the industry. All manufacturers of semiconductor products are experiencing counterfeiting of their parts. Customers who inadvertently purchase counterfeit parts experience many problems such as loss of brand reputation, substandard performance, failed applications, and increased cost of production and manufacturing delays. Fairchild is taking strong measures to protect ourselves and our customers from the proliferation of counterfeit parts. Fairchild strongly encourages customers to purchase Fairchild parts either directly from Fairchild or from Authorized Fairchild Distributors who are listed by country on our web page cited above. Products customers buy either from Fairchild directly or from Authorized Fairchild Distributors are genuine parts, have full traceability, meet Fairchild's quality standards for handling and storage and provide access to Fairchild's full range of up-to-date technical and product information. Fairchild and our Authorized Distributors will stand behind all warranties and will appropriately address any warranty issues that may arise. Fairchild will not provide any warranty coverage or other assistance for parts bought from Unauthorized Sources. Fairchild is committed to combat this global problem and encourage our customers to do their part in stopping this practice by buying direct or from authorized distributors.

PRODUCT STATUS DEFINITIONS

Definition of Terms

Datasheet Identification	Product Status	Definition
Advance Information	Formative / In Design	Datasheet contains the design specifications for product development. Specifications may change in any manner without notice.
Preliminary	First Production	Datasheet contains preliminary data; supplementary data will be published at a later date. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve design.
No Identification Needed	Full Production	Datasheet contains final specifications. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve the design.
Obsolete	Not In Production	Datasheet contains specifications on a product that is discontinued by Fairchild Semiconductor. The datasheet is for reference information only.

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